DualVGR: A Dual-Visual Graph Reasoning Unit for Video Questioning Answering

Jianyu Wang[1], Bing-Kun Bao[1], Changsheng Xu[2]

[1]Nanjing University of Posts and Telecommunications, Nanjing, China.

[2]Institute of Automation, Chinese Academy of Science, Beijing, China.

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Motivation:

➢ Despite GNN has arouse many research interests in VideoQA, existing graph-based methods neglect two properties of VideoQA:

1. Different questions may require different amount of video clips or objects to infer the answer with relational reasoning;
2. Appearance and motion features have complicated interdependence which are correlated and complementary to each other.

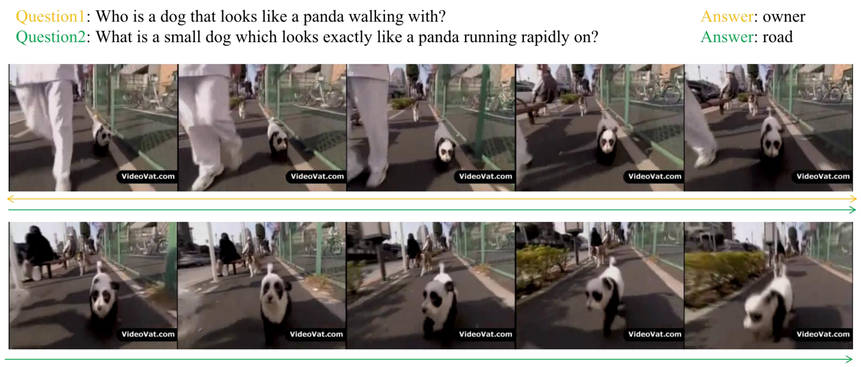

Figure 1. Answering questions require agents to utilize both appearance and motion information, and usually requires them to be capable of multiple steps of spatio-temporal reasoning.

Contributions:

➢ DualVGR Unit includes two components. One is an explainable Query Punishment Module to filter out irrelevant visual-based information. The other is a Video-based Multi-view Graph Attention Network to perform spatial-temporal relational reasoning such that the relations between appearance and motion can be adaptively captured;

➢ We incorporate our DualVGR unit into a full DualVGR network by stacking it in an iterative manner. Through multi-step reasoning, our method achieves state-of-the-art or competitive results on several mainstream datasets: MSVD-QA , MSRVTT-QA and SVQA .

Project Page

The proposed DualVGR

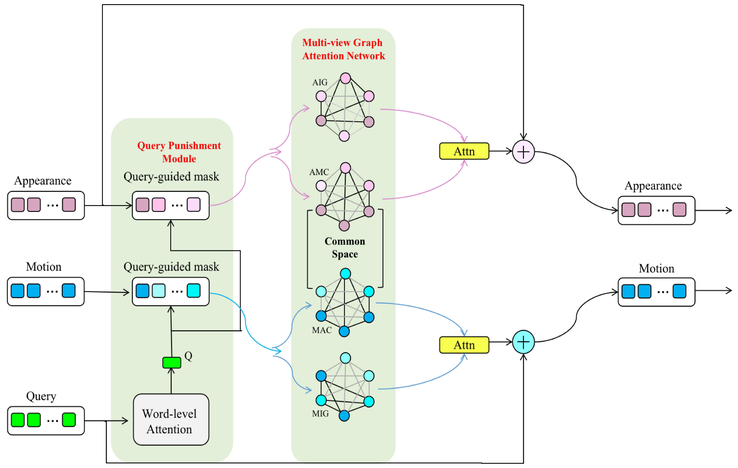

Figure 2. Our DualVGR efficiently represents videos as amalgam of complementing information, including appearance, motion and relation features.

➢ Component1: The goal of Query Punishment Module is to generate the masked values to filter out irrelevant video clips:

$$M_i^{(t)} = V_i^{(t)} W^{(t)} q^{(t)}$$

➢ Then we obtain each clip's feature by multiplying it with corresponding mask value:

$$V_i^{(t+1)} = M_i^{(t)} V_i^{(t)}$$

➢ Component2: Multi-view Graph Attention Network is proposed to both consider the relations among video clips within each channel as well as the relations between two channels for each video clip.

1) Specific Embedding to capture relations within each visual channel: Embedding similarity matrix for consistency constraint.

2) common embedding to capture the relations between these two channels: Hilbert-Schmidt Independence Criterion(HSIC) for disparity constraint.

Experimental Result

Table 1. Comparison to the SOTA on MSVD-QA and MSRVTT-QA

	MSVD-QA						MSRVTT-QA					
	what	who	how	when	where	ALL	what	who	how	when	where	ALL
HME [14]	22.4	50.1	73.0	70.7	42.9	33.7	26.5	43.6	82.4	76.0	28.6	33.0
HGA [18]	23.5	50.4	83.0	72.4	46.4	34.7	29.2	45.7	83.5	75.2	34.0	35.5
HCRN [15]	/	/	/	/	/	36.1	/	/	/	/	/	**35.6**
TSN [41]	25.0	51.3	**83.8**	**78.4**	**59.1**	36.7	27.9	**46.1**	**84.1**	**77.8**	**37.6**	35.4
Ours	**28.65**	**53.82**	80.00	70.69	46.43	**39.03**	**29.40**	45.56	79.76	76.66	36.40	35.52

Table 2. Ablation Results

Model	Step T	SVQA	MSVD-QA					
			what	who	how	when	where	ALL
Two-stream								
AG	$T = 1$	46.54	26.97	52.17	79.46	**74.14**	**50.00**	37.42
MG	$T = 1$	48.19	23.11	49.56	77.30	70.69	46.43	34.04
FG	$T = 1$	48.45	27.77	51.85	**82.97**	67.24	**50.00**	37.87
Multi-view								
Bigraph	$T = 1$	47.53	26.09	53.10	77.57	70.69	35.71	37.10
MVgraph	$T = 1$	47.59	26.75	52.07	75.68	72.41	46.43	37.13
Punishment								
PFG	$T = 1$	48.96	27.50	53.87	74.59	67.24	46.43	38.16
shareDVGR	$T = 1$	48.96	27.94	52.35	77.84	72.41	**50.00**	38.03
DualVGR	$T = 1$	**49.12**	**28.65**	**53.82**	80.00	70.69	46.43	**39.03**
self-attention								
simpleDualVGR	$T = 4$	49.02	/	/	/	/	/	/
DualVGR	$T = 4$	50.38	/	/	/	/	/	/

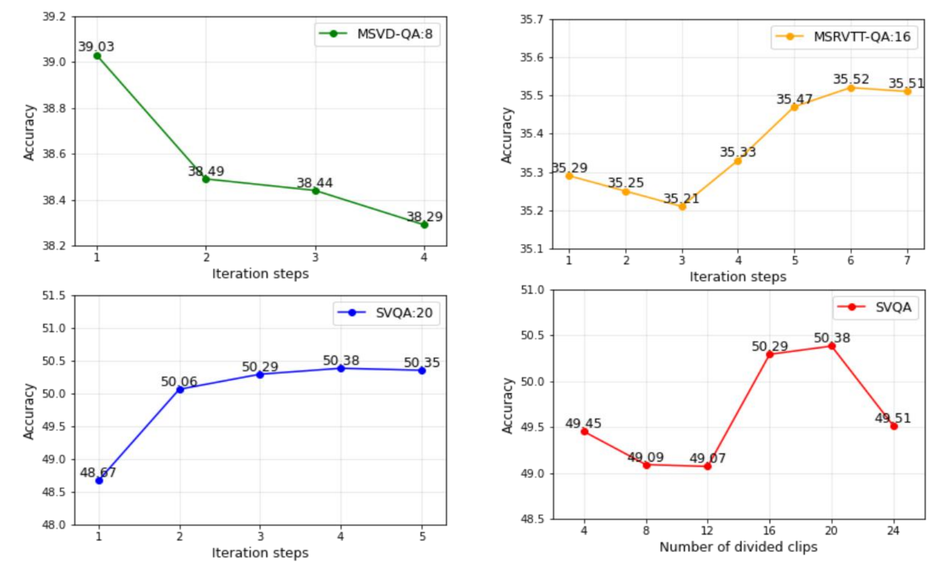

Figure 3. Detailed Analysis of the number of iteration steps and divided clips.

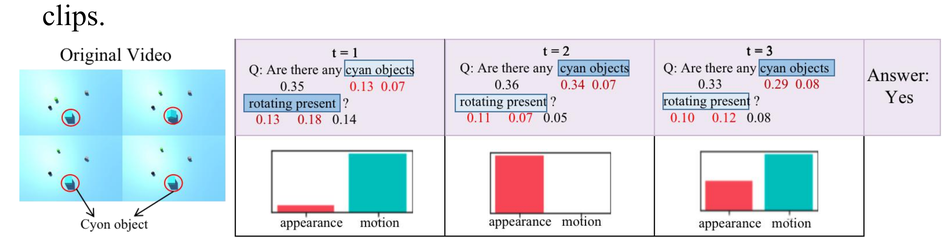

Figure 4. Visualization of the attention scores in each step's query representation and the generated query-guided visual mask of each visual feature in each step on SVQA.